LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 6 Test: Control and Coordination

Chapter / Topic: Control and Coordination

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following is a plant hormone? **[1]**

(a) Insulin (b) Cytokinin
(c) Thyroxin (d) Oestrogen

Q2. The gap between two neurons is called a: **[1]**

(a) dendrite (b) axon
(c) synapse (d) impulse

Q3. The brain is responsible for: **[1]**

(a) thinking (b) regulating the heartbeat
(c) balancing the body (d) all of the above

Q4. Which hormone regulates the blood sugar level? **[1]**

(a) Insulin (b) Adrenaline
(c) Thyroxin (d) Growth hormone

Q5. The bending of a plant shoot towards light is called: **[1]**

(a) geotropism (b) hydrotropism
(c) chemotropism (d) phototropism

Q6. **Assertion (A):** Iodised salt is advised in our diet. **Reason (R):** Iodine is needed by the thyroid gland to make thyroxin. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What is a reflex action? Give two examples. **[2]**

OR

How do the leaves of the touch-me-not plant respond to touch? How is this movement different from a shoot bending towards light?

Q8. Why is adrenaline called the 'emergency hormone'? Give two of its effects on the body. **[2]**

SECTION C

6 marks

Q9. Answer the following about a neuron: **[3]**

(i) In which part of a neuron is information first acquired? **[1]**

(ii) Through which part does the information travel as an electrical impulse? **[1]**

(iii) How is the impulse passed on to the next neuron? **[1]**

Q10. Name the plant hormone that: **[3]**

(i) promotes cell division. **[1]**

(ii) inhibits growth and causes wilting of leaves. **[1]**

(iii) is made at the shoot tip and helps cells grow longer. **[1]**

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) For each condition, name the gland and the hormone involved: (a) dwarfism, (b) diabetes, (c) goitre. **[3]**

(ii) Why is a feedback mechanism important in the secretion of hormones? Give an example. **[2]**

OR

(a) Name the three main regions of the brain and give one function of each. [3]
(b) How is the brain protected? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A student placed a potted plant near a window for a week. The shoot bent towards the window, while the roots grew downwards. The student read that a hormone made at the shoot tip diffuses towards the shaded side of the shoot. This makes the cells on that side grow longer, so the shoot bends towards light. **[4]**

(i) Name the response of the shoot towards light. **[1]**

(ii) Name the hormone involved, and say where it is made. **[1]**

(iii) Explain why the shoot bends towards light. Name the response of the roots to gravity. **[2]**

~ All the best! ~